

Incorporating RNA Modifications into ViennaRNA's Predictive Models

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Hinxton, 2 April 2025



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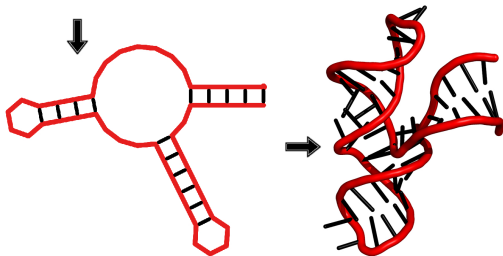


RNA Forecast

The RNA Folding Problem

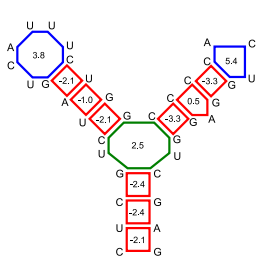
Hierarchical folding: Secondary structure forms first then helices arrange to form tertiary structure

GCGCUCUGAUGAGGCCGCAAGGCCGAAACUGCCGCAAGGCAGUCAGCGC



- Secondary structure captures the majority of stabilizing
- Convenient and biologically useful description
- Efficient algorithms for many thermodynamic properties
- High prediction accuracy for small RNAs
- Good basis for 3D structure modeling and RNA folding kinetics

RNA Secondary Structures - Nearest Neighbor Energy Model



$$E(\text{Y}) = U_{C \rightarrow G}^A \begin{matrix} G \\ -2.1 \\ C \end{matrix} + U_{U \rightarrow A}^C \begin{matrix} C \\ -2.4 \\ A \end{matrix} + U_{G \rightarrow G}^G \begin{matrix} G \\ -2.4 \\ G \end{matrix} + U_{U \rightarrow U}^G \begin{matrix} G & C \\ & 2.5 \\ G & C \end{matrix} + U_{U \rightarrow C}^U \begin{matrix} G \\ -2.1 \\ C \end{matrix} + U_{A \rightarrow G}^U \begin{matrix} U \\ -1.0 \\ U \end{matrix} + U_{G \rightarrow A}^C \begin{matrix} C \\ -2.1 \\ A \end{matrix} + U_{A \rightarrow U}^A \begin{matrix} U & C \\ & 3.8 \\ U & G \end{matrix} + U_{C \rightarrow G}^C \begin{matrix} C \\ -3.3 \\ G \end{matrix} + U_{C \rightarrow G}^C \begin{matrix} C \\ 0.5 \\ G \end{matrix} + U_{C \rightarrow G}^C \begin{matrix} C \\ -3.3 \\ G \end{matrix} + U_{A \rightarrow U}^A \begin{matrix} A & C \\ & 5.4 \\ G & U \end{matrix}$$

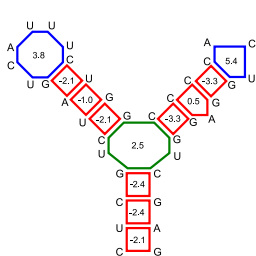
$$= -6.50 \text{ kcal/mol}$$

- Decomposition of Secondary structures s into loops L
- Contributions of a base pair depend on neighboring pairs
- Each loop L is assigned a free energy contribution $E_L^{[1]}$

$$E(s) \approx \sum_{L \in s} E_L$$

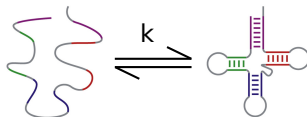
[1] Mittal et al., "NNDB: An Expanded Database of Nearest Neighbor Parameters for Predicting Stability of Nucleic Acid Secondary Structures", 2024, J. Mol. Biol., 168549

RNA Secondary Structures - Nearest Neighbor Energy Model



$$E(\text{Y}) = U_{C \text{ G}}^{\text{A}}(-2.1) + U_{U \text{ A}}^{\text{C}}(-2.4) + U_{G \text{ C}}^{\text{U}}(-2.4) + U_{G \text{ U}}^{\text{C}}(2.5) + U_{U \text{ G}}^{\text{A}}(-2.1) + U_{A \text{ U}}^{\text{C}}(3.8) + U_{C \text{ A}}^{\text{G}}(-3.3) + U_{G \text{ A}}^{\text{C}}(0.5) + U_{G \text{ C}}^{\text{A}}(-3.3) + U_{C \text{ U}}^{\text{A}}(5.4)$$

$$= -6.50 \text{ kcal/mol}$$



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$$\Delta G = G(\textit{folded}) - G(\textit{unfolded})$$

$$\Delta G = \Delta H - T\Delta S$$

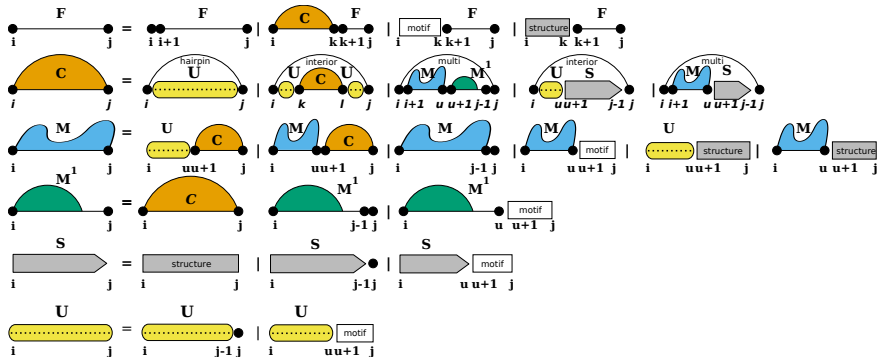
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RNA Secondary Structure Prediction

Nussinov decomposition scheme:



Full decomposition scheme:

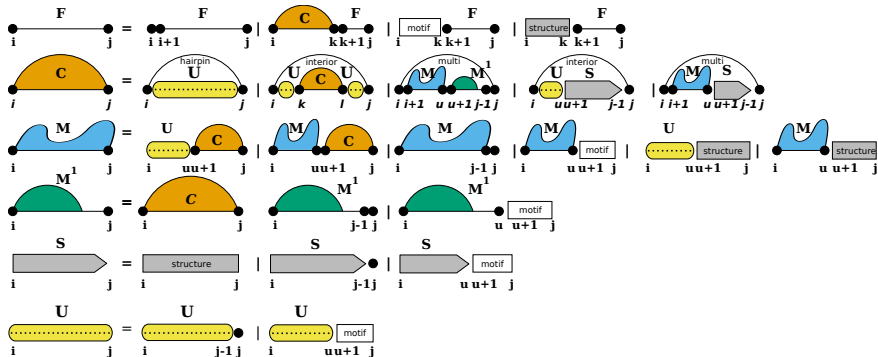


RNA Secondary Structure Prediction

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Full decomposition scheme:



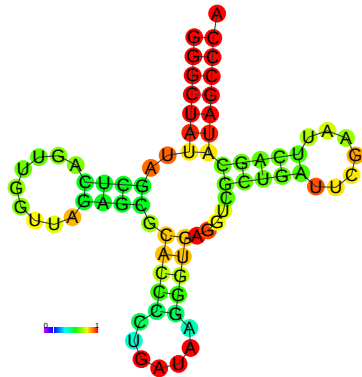
Availability

Implemented in ViennaRNA Package including libRNA and scripting language interface

ViennaRNA **Secondary Structure Prediction**

- Minimum free energy (MFE) structure

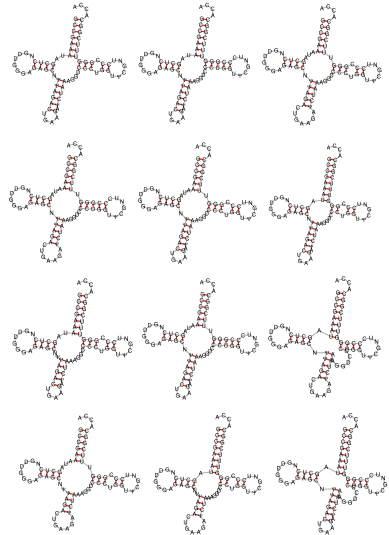
MFE structure



ViennaRNA Secondary Structure Prediction

- Minimum free energy (MFE) structure
- Suboptimal secondary structures

suboptimal secondary structures



ViennaRNA Secondary Structure Prediction

- Minimum free energy (MFE) structure
- Suboptimal secondary structures
- Partition function Z

$$p(s) \propto e^{-\beta E(s)} \quad \text{with} \quad \beta = \frac{1}{kT}$$

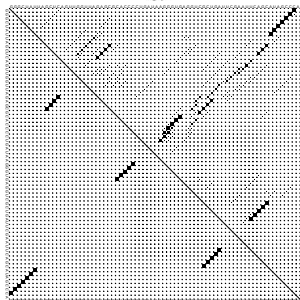
$$Z = \sum_s e^{-\beta E(s)} \equiv \sum_E g(E) e^{-\beta E}$$

- Probability $p(\mathcal{F})$ of feature $\mathcal{F}$

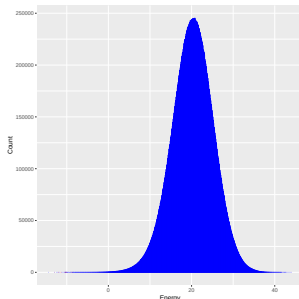
$$p(\mathcal{F}) = \frac{Z_{\mathcal{F}}}{Z} \quad \text{with} \quad Z_{\mathcal{F}} = \sum_{s|\mathcal{F} \in s} e^{-\beta E(s)}$$

- Base Pair Probabilities, Accessibility, etc.

base pair probability dot plot



density of states (DoS)



ViennaRNA Secondary Structure Prediction

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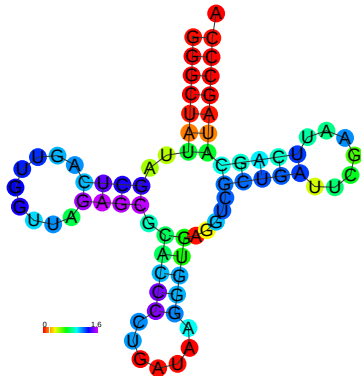
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- Base Pair Probabilities, Accessibility, etc.
- Global / local reliability measures

positional ('Shannon') entropy



$$PE(i) = - \sum_k p_{ik} \ln p_{ik}$$

ViennaRNA Secondary Structure Prediction

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- Base Pair Probabilities, Accessibility, etc.
- Global / local reliability measures
- **Hard constraints** limit candidate search space
- **Soft constraints** change candidate evaluation

standard NN energy model

$$E(s) \approx \sum_{L \in s} E_L$$

soft constraints allow for adding pseudo energy terms $E_L^{m_i}$

$$E(s) = \sum_{L \in s} E_L + \Delta \Delta E_L^m$$

$$\Delta \Delta E_L^m = \sum_i (E_L^{m_i} - E_L)$$

ViennaRNA Secondary Structure Prediction

- Minimum free energy (MFE) structure
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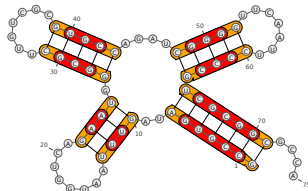
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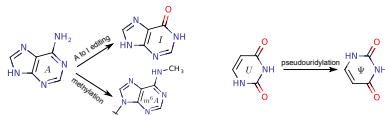
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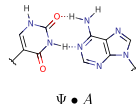
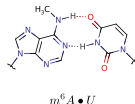
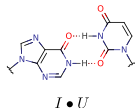
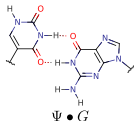
$$\Delta G(i) = m * \ln(\text{reactivity}[i] + 1) + b$$

Modified Bases in RNA

Post-transcriptional RNA modifications (epitranscriptome):

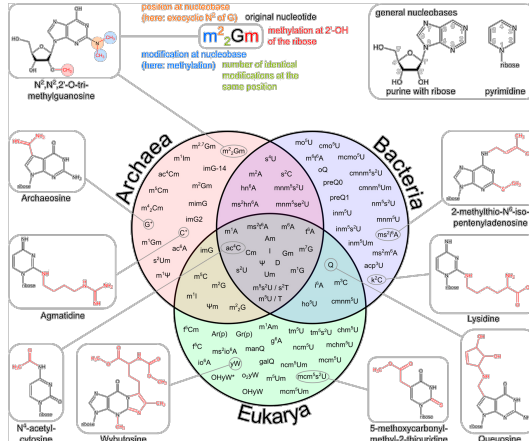


- Modomics Database^[2] lists 335 different modified bases
- Commonly known modifications: *I*, *Ψ*, *m*⁶*A*, *m*¹*A*, *m*⁵*C*, ...
- Function and purpose of modifications still largely unknown
- Structural effects of base modifications:
 - correct folding of ncRNAs into functional structures (tRNA, rRNA, etc.)
 - regulation of protein binding sites (mRNAs, lncRNAs)
 - regulation of RNA-RNA binding sites (siRNA, miRNA)
 - Modifications may change pairing partner preference
 - Modifications may (de-)stabilize loop formation



[2] Cappannini et al., "MODOMICS: a database of RNA modifications and related information.", 2024, NAR 52.D1, D239–D244

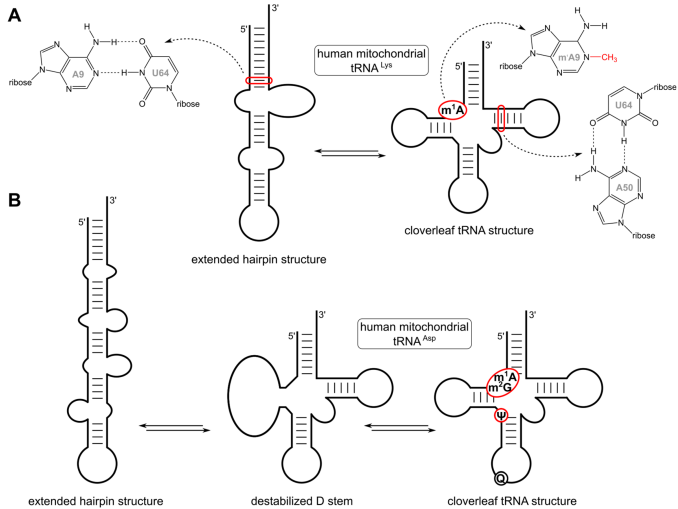
93 known post-transcriptional modifications



- Modifications can be subtle from the RNA structure perspective
- Some are essential to induce structural domain rearrangements

[3] Lorenz, Lünse, and Mörl, “tRNA Modifications: Impact on Structure and Thermal Adaptation”, *Biomolecules* 2017, 7, 35

Modifications in tRNA^[3]



RNA Secondary Structure Prediction and Modified Bases

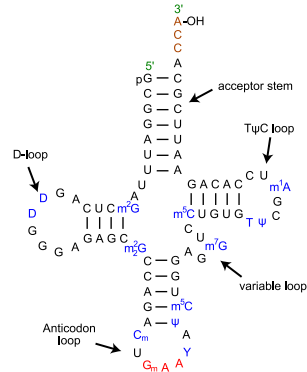
How to model modified bases in prediction algorithms?

Actual Requirements:

- Enhanced Nucleotide Alphabet
- Additional base pairing rules
- Corresponding energy parameters

Obstacles:

- 2D structure effects are known only for a few modifications
- 3D effects either are unknown or impossible to model
- Combinatorial explosion for energy parameters / pairing rules



RNA Modifications - Status Quo of the ViennaRNA Package

Built-in modifications:

- inosine (*I*)
- pseudouridine (Ψ)
- m^6A
- m^5C
- *7DA*
- purine (a.k.a. nebularine)
- dihydrouridine (*D*)

Available through JSON files:

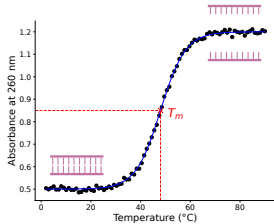
- $m^1\Psi$
- m^5C

Prevent base pairing (hard constraints):

- m^1A
- m^1G
- m_2^2G
- m^3U
- m_2^6A
- $acp^3\Psi$

Energy parameters are incomplete and mostly restricted to base pair stacking

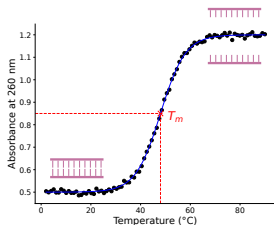
How To Obtain Energy Parameters for Modified Bases?



Experiments

- UV melting approach
- expensive and time-consuming
- Availability of modifications can be a problem

How To Obtain Energy Parameters for Modified Bases?

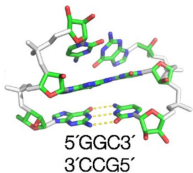


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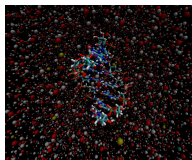
Computational methods

Coarse grained modeling



- No solvent
- Less degrees of freedom
- Lower computational costs

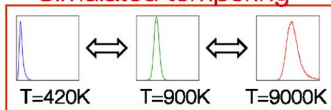
Molecular dynamics



- Solvent
- Higher computational costs
- Fewer approximations
- Complex to set-up

RECCES^[4] Simulation

Simulated tempering



WHAM



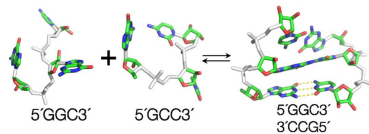
Density of states

$$Z = \sum_{s \in \Omega} e^{-\beta E(s)}$$

ΔG ΔH

- Monte Carlo simulation
- Temperature changes during the simulation
 - To overcome energy barriers
- Energy function
 - Combination of Physics- and knowledge-based terms

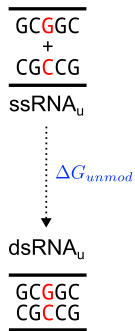
$$\Delta G = G(\text{folded}) - G(\text{unfolded})$$



$$\Delta G \left(\begin{smallmatrix} 5'GGC \\ 3'CCG \end{smallmatrix} \right) = G \left(\begin{smallmatrix} 5'GGC \\ 3'CCG \end{smallmatrix} \right) - G(\text{GGC}) - G(\text{CCG})$$

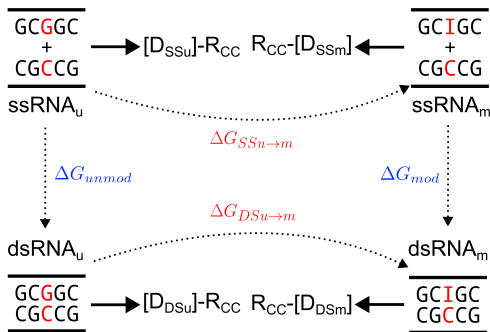
[4] Chou et al., "Blind tests of RNA nearest-neighbor energy prediction", 2016 Proceedings of the National Academy of Sciences, 113(30), 8430-8435.

Transformato^[5] Thermodynamic Cycle



[5] Karwounopoulos et al. "Relative binding free energy calculations with transformato: A molecular dynamics engine-independent tool." *Frontiers in Molecular Biosciences* 9 (2022)

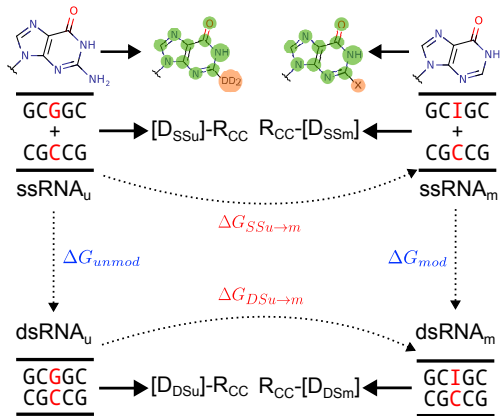
Transformato^[5] Thermodynamic Cycle



- Construction of an alchemical path

$$\begin{aligned} \Delta\Delta G_{unmod \rightarrow mod} &= \Delta G_{mod} - \Delta G_{unmod} \\ &= \Delta G_{DSu \rightarrow m} - \Delta G_{SSu \rightarrow Sm} \end{aligned}$$

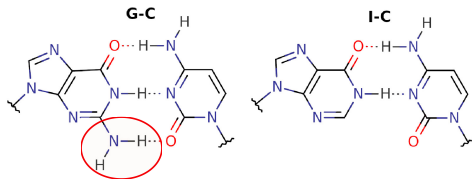
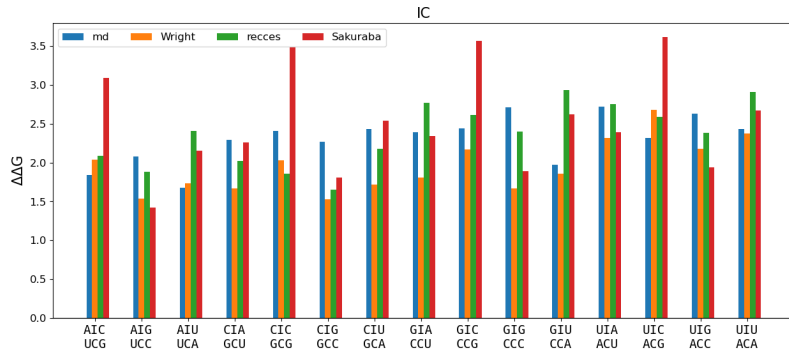
Transformato^[5] Thermodynamic Cycle



- Construction of an alchemical path
- Maximum common substructure

$$\begin{aligned} \Delta \Delta G_{\text{unmod} \rightarrow \text{mod}} &= \Delta G_{\text{mod}} - \Delta G_{\text{unmod}} \\ &= \Delta G_{\text{DSu} \rightarrow \text{Sm}} - \Delta G_{\text{SSu} \rightarrow \text{Sm}} \end{aligned}$$

GC - IC



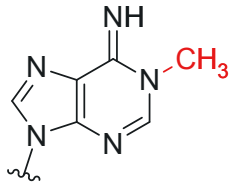
Aedes albopictus Mitochondrial tRNA^{Asp}

AAAAAAUU"GUUUAUCAAACCPAGUAUGUC6AACUAAAAAAUUA GAUCAUCUAAUA PPUUUUACCA
 AAAAAAUUAGUUUAUCAAACCUUAGUAUGUCA AACUAAAAAAUUA GAUCAUCUAAUAUUUUUUACCA

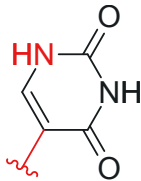
": 1-methyladenosine

P: pseudouridine

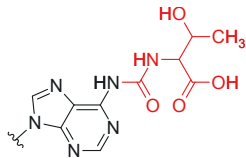
6: N6-threonylcarbamoyladenine



*m*¹A



Ψ



*t*⁶A

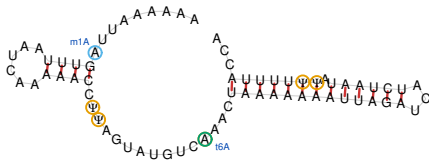
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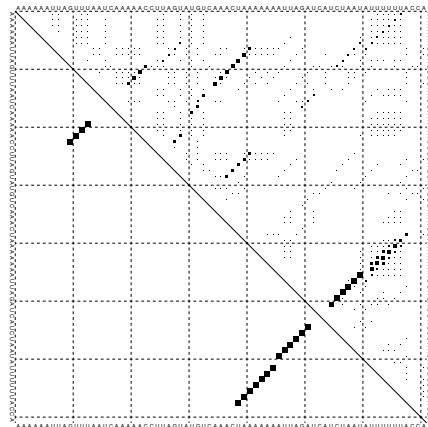
": 1-methyladenosine

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- Known parameters
- Prevent base pairing
- Unknown parameters



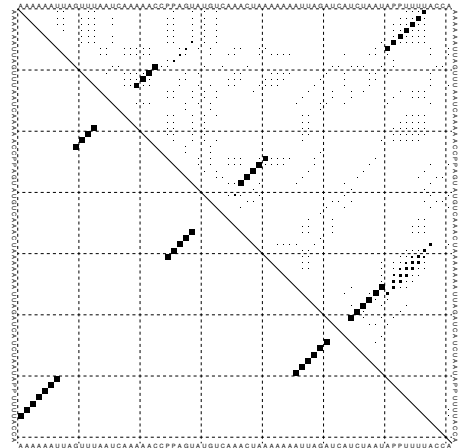
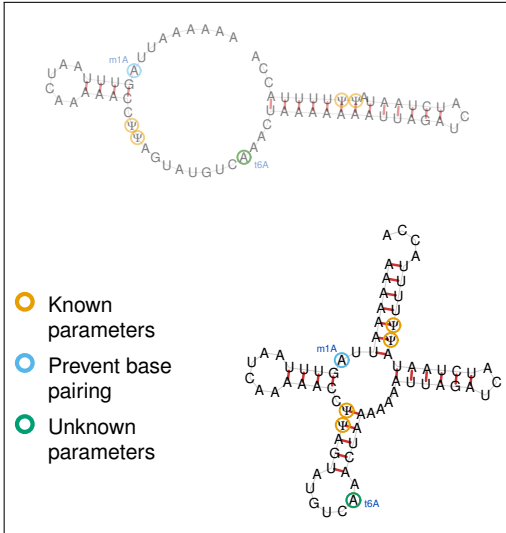
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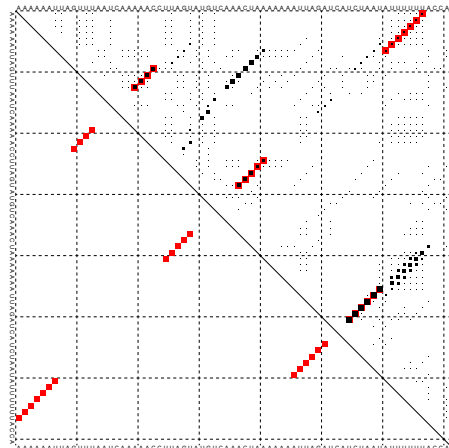
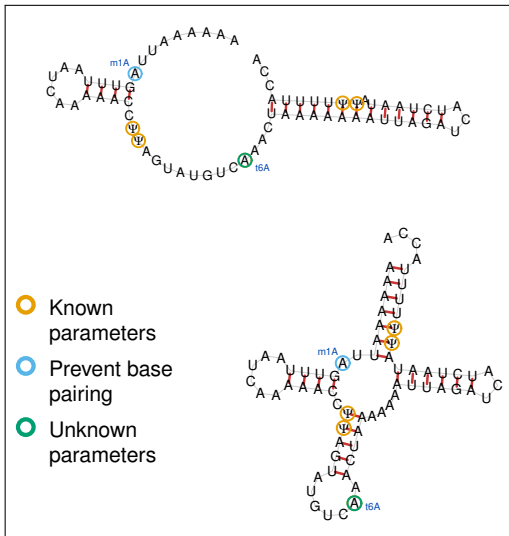
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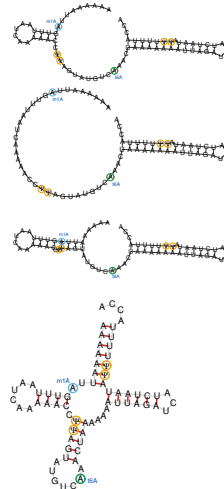
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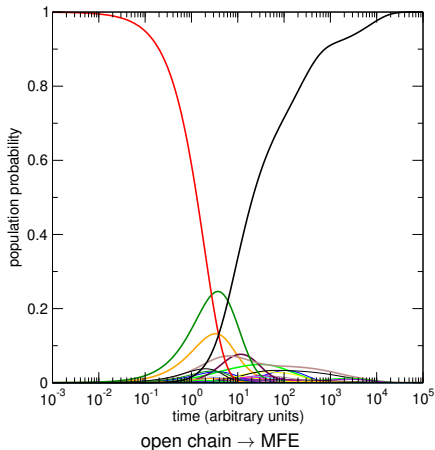
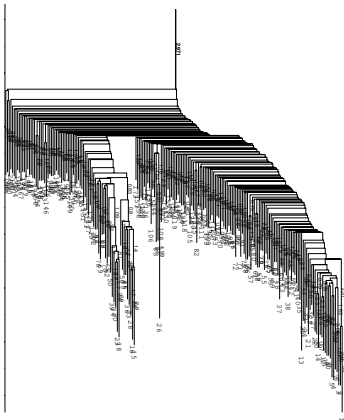
AAAAAAUAGUUUAUCAAACCUUAGUAUGUCAAACUAAAAAAUAGAUAUCUAAUAUUUUUACCA -5.10
(((.....))).....((((((((((((.....)))))).))))))... -5.10
(((.....))).....((((((((((((.....)))))).))))))... -5.00
((((((((((((.....)))))).)))))).....((((((((((((.....)))))).))))))... -4.60
(((.....))).....((((((((((((.....)))))).))))))... -4.40
(((.....))).....((((((((((((.....)))))).))))))... -4.30
 (((((((.....((((.....))))).((((.....))))).((((.....)))))))))... -4.10
(((.....))).....((((((((((((.....)))))).))))))... -4.00
((((((((((((.....)))))).)))))).....((((((((((((.....)))))).))))))... -4.00
((((((((.....((((.....))))).((((.....))))).((((.....)))))))))... -4.00
(((.....))).....((((((((((((.....)))))).))))))... -3.90

suboptimal of unmodified tRNA



Aedes albopictus Mitochondrial tRNA^{Asp} Folding Dynamics

Energy landscape^[6] and folding dynamics^[7] of tRNA^{Asp} **with modifications**

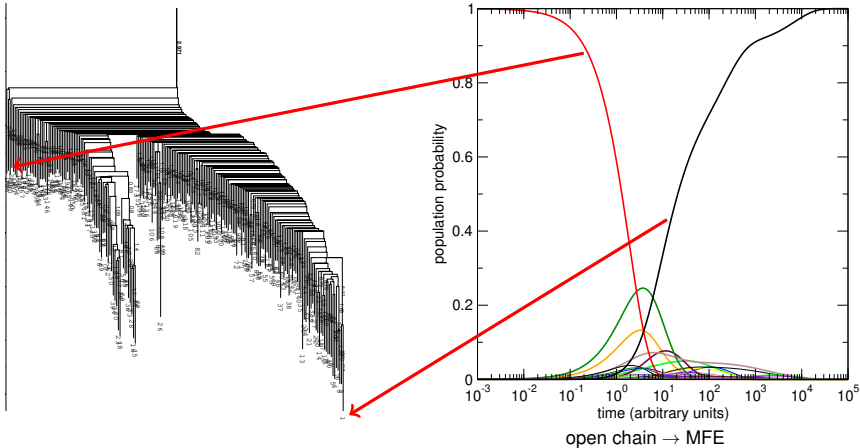


[6] Flamm et al., "Barrier trees of degenerate landscapes", 2002, Z. Phys. Chem. 216: 155-73

[7] Wolfinger et al., "Efficient Computation of RNA Folding Dynamics", 2004, J.Phys.A: Math.Gen. 37(17):4731-41

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Energy landscape^[6] and folding dynamics^[7] of tRNA^{Asp} **with modifications**



[6] Flamm et al., "Barrier trees of degenerate landscapes", 2002, Z. Phys. Chem. 216: 155-73

[7] Wolfinger et al., "Efficient Computation of RNA Folding Dynamics", 2004, J.Phys.A: Math.Gen. 37(17):4731-41

Modified Bases Significantly Influence Structure Space

Takeaway Message:

- We employ MC and MD approaches to compute missing energy parameters
- Additional parameters do not necessarily increase performance
- Constraints become complex for more modifications and contexts
- Unrealistic to include full parameters with many modified bases
- Base annotation is not unique (tRNAdb^[8], RNAmo^[9], MODOMICS^[10])

ViennaRNA Package:

- Can handle RNA modifications via the soft constraints framework
- New parameters - even partial - can easily be added as they become available
- Software is freely available for academic and commercial use

[8] Jühling et al., "tRNAdb 2009: compilation of tRNA sequences and tRNA genes.", 2009, NAR 37, D159–D162

[9] Liu et al., "RNAmo: an integrated system for the annotation of mRNA modifications", 2019, NAR 47, W548–W555

[10] Boccaletto et al., "MODOMICS: a database of RNA modification pathways. 2017 update", 2018, NAR 46, D303–D307

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